

Reading the step diagnostics

A companion to `step_diagnostics_v1.ipynb`

Abstract

We explain the eight columns of the step-diagnostic tables and the six functions that produce them, and we interpret the tables of `step_diagnostics_v1.ipynb`. Truncated conjugate gradient leaves its loop in three ways and every one of them returns a positive multiple of $-g_k$, so a one-dimensional Krylov subspace forces the cosine test to pass. The two Cauchy statistics are therefore nested rather than independent, `cg1` $\leq$ `cauchy`, which is why they print the same number on 28 of 30 rows. On the two rows where they separate the accepted explanation, that g_k is near an eigenvector, is refuted by the measurement. The quantity that separates the populations is the gap between the first conjugate-gradient iterate and the boundary, whose median is 6.5×10^{-9} at the disagreeing iterations and 1.6×10^{-3} at the agreeing ones. We give the arithmetic that turns that gap into an angle below the 10^{-12} threshold, and we check it. The truncation condition is confirmed on 17755 of 17763 exact-Hessian iterations, and all eight exceptions are the one conjugate-gradient exit the condition does not model.

1 The setting

The test function is

$$f(p) = p_1^4 - p_1^3 + \left(\frac{1}{4} - \frac{p_1}{2}\right)p_2^2 + \frac{p_2^4}{4}, \quad (1)$$

with four critical points. The origin is degenerate, $\nabla^2 f(0,0)$ being singular and positive semi-definite. The point $(0.75,0)$ is a strict saddle with $\lambda_{\min} = -1/4$. Two local minimisers sit at $\left(\frac{1+\sqrt{5}}{4}, \pm\sqrt{\frac{\sqrt{5}-1}{4}}\right)$.

Every run in the notebook uses `R-grad` capped, so $\Delta_k = \mu_k \|g_k\|$ with $\mu_k \leq \bar{\mu}$, sweeping $\bar{\mu}$ over $\{0.01, 0.05, 0.1, 0.3, 0.5, 1, 2, 4, 8, 128\}$ against three model Hessians, `ExactHessian`, `SR1` and `LBFGS`. The subproblem solver is truncated conjugate gradient throughout, apart from one comparison against `ExactMS`. The starting point is $(-0.5, 0.6)$ and the tolerance 10^{-9} , with at most 2000 iterations.

Everything below is read from `stats.solver_specific`. Nothing is recomputed from the iterates that the trace already carries.

1.1 The three ways truncated conjugate gradient stops

The whole of Section 2 follows from the recurrence, so it is worth setting down first. `SteihaugCG` starts at $s = 0$ with residual $r_0 = -g_k$ and first direction $d_0 = -g_k$, and it leaves the loop in exactly three ways.

- (E1) *Negative curvature.* If $d_0^\top B_k d_0 \leq 0$ the recurrence runs to the boundary along d_0 and returns $s_k = -\Delta_k g_k / \|g_k\|$.
- (E2) *The boundary.* Otherwise it forms $\alpha_0 = \|g_k\|^2 / (g_k^\top B_k g_k)$ and the candidate $z_1 = -\alpha_0 g_k$. If $\|z_1\| \geq \Delta_k$ it truncates to the boundary and returns $s_k = -\Delta_k g_k / \|g_k\|$.
- (E3) *The forcing condition.* Otherwise it accepts z_1 , and if the new residual satisfies $\|r_1\| \leq \min\{\chi, \|g_k\|^\theta\} \|g_k\|$ it stops there, returning $s_k = z_1 = -\alpha_0 g_k$.

Only if none of the three fires does the recurrence build a second direction.

The consequence used throughout is immediate. *In all three exits the returned step is a positive multiple of $-g_k$.* A run that leaves after one conjugate-gradient iteration has taken a step along the steepest descent direction, whatever the model Hessian was.

2 The six columns of Table 1

2.1 trunc, inside and krylov

These are the three step classes of `step_classes`, as fractions of the iterations that happened. Write c_k for the inner iteration count and a_k for the activity flag, which is true when $\|s_k\| = \Delta_k$.

$$\text{trunc} = \frac{1}{K} \#\{k : c_k = 1 \text{ and } a_k\}, \quad (2)$$

$$\text{inside} = \frac{1}{K} \#\{k : c_k = 1 \text{ and } \neg a_k\}, \quad (3)$$

$$\text{krylov} = \frac{1}{K} \#\{k : c_k \geq 2\}. \quad (4)$$

The three partition the iterations, so they sum to one.

`trunc` collects (E1) and (E2). The step is $-\Delta_k g_k / \|g_k\|$, so the model Hessian has touched neither the direction nor the length: the direction is $-g_k$ and the length is Δ_k . **This is the only one of the three that is gradient descent in disguise**, and the step length is the radius rather than anything the model chose.

`inside` is (E3). The direction is again $-g_k$, but the length is $\alpha_0 \|g_k\| = \|g_k\|^3 / (g_k^\top B_k g_k)$, which is the exact minimiser of the model along $-g_k$. The Hessian set the length and not the direction. This is the Cauchy point in the strict sense.

`krylov` is everything else, where a second direction entered and the Hessian influenced the direction as well.

Interpretation. Read `trunc` as the fraction of the run on which the trust region, rather than the model, decided the step. A `trunc` of one means the method was gradient descent with a prescribed step length for the whole run. `krylov` near one means the model was doing the work.

`step_classes` returns (NaN, NaN, NaN) under `ExactMS`, which reports $c_k = 0$ throughout, since a direct solve has no inner iteration count. The notebook confirms this: `cg_iters` takes the values $\{2, 1\}$ under `SteihaugCG` and $\{0\}$ under `ExactMS`, and the classes are (0.999, 0.000, 0.001) and (NaN, NaN, NaN) respectively.

2.2 cauchy

The cosine-based fraction reported by `hypotheses_report`,

$$\text{cauchy} = \frac{1}{K} \#\left\{k : |\cos(s_k, -g_k) - 1| \leq 10^{-12}\right\}, \quad \cos(s_k, -g_k) = \frac{-g_k^\top s_k}{\|g_k\| \|s_k\|}. \quad (5)$$

It measures how often the step *pointed* along steepest descent, to within 10^{-12} . It says nothing about the length.

2.3 cg1

The Krylov dimension,

$$\text{cg1} = \frac{1}{K} \#\{k : c_k = 1\} = \text{trunc} + \text{inside}. \quad (6)$$

It measures how often the recurrence stopped before building a second direction. The identity with the two step classes is exact, not approximate, since (2) and (3) split $\{c_k = 1\}$ on the activity flag.

2.4 `cos_min`

The minimum of $\cos(s_k, -g_k)$ over the run,

$$\text{cos_min} = \min_k \cos(s_k, -g_k). \quad (7)$$

This is the column that stops a fraction from being read as a description of the whole run. `cauchy` says how *often* the step was Cauchy; `cos_min` says how far from Cauchy the run ever got. A run with `cos_min` = 1 was gradient descent at *every* iteration, not at most of them.

Read *cos_min* rather than the cosine at the final iterate. The last value is one number out of however many iterations ran, and the two disagree: in the companion notebook `exact_steihaug_v1` the final-iterate cosine is 1.00000 at $\bar{\mu} = 4$ while `cauchy` is 0.4885 there.

3 Why `cauchy` and `cg1` answer the same question

Both ask whether the model Hessian influenced the *direction* of the step. They ask it of different objects. `cg1` asks it of the Krylov subspace, $\dim \mathcal{K}_k = 1$, and `cauchy` asks it of the angle the step came out at.

The two are linked in one direction by Section 1.1.

Proposition 1 (the inclusion). *For every k with $c_k = 1$, the returned step satisfies $s_k = -tg_k$ for some $t > 0$, hence $\cos(s_k, -g_k) = 1$ exactly. Consequently*

$$\{k : c_k = 1\} \subseteq \{k : |\cos(s_k, -g_k) - 1| \leq 10^{-12}\}, \quad \text{so} \quad \text{cg1} \leq \text{cauchy}. \quad (8)$$

Proof. Each of (E1), (E2) and (E3) returns a positive multiple of $d_0 = -g_k$, the first with $t = \Delta_k / \|g_k\|$, the second likewise, the third with $t = \alpha_0 > 0$. $\square$

So `cauchy` counts everything `cg1` counts, and possibly more. The inequality (8) is the whole relationship, and the two agree whenever the extra set is empty.

3.1 Why they hold the same value on nearly every row

Equality fails only at an iteration with $c_k \geq 2$ whose returned step is nonetheless within 10^{-12} of the $-g_k$ direction. That asks the second and later conjugate-gradient directions to add nothing *measurable* to the direction, while still being taken.

That is a demanding requirement, and on this problem it is met on 2 of the 30 rows of Table 1. On the remaining 28 the extra set is empty and the two columns print the same number, which is not a coincidence but Proposition 1 with nothing left over.

3.2 The two rows where they differ, and why

Under `ExactHessian` at $\bar{\mu} = 4$ and $\bar{\mu} = 8$ the columns separate, 0.4885 against 0.3345 and 0.4815 against 0.3319. Every one of the 308 and 251 disagreeing iterations goes the same way, the cosine counting an iteration the Krylov dimension does not, exactly as (8) requires.

The natural explanation is that g_k is close to an eigenvector of H_k there, so the second direction adds nothing. **It does not survive the measurement.** Pooled over the disagreeing runs, `eig_deviation` has median 7.0695×10^{-1} at the disagreeing iterations against 7.0617×10^{-1} at the agreeing ones. The two populations are indistinguishable on that statistic, and both are far from zero.

What separates them is the boundary. Writing $\varepsilon_k = 1 - \|z_1\| / \Delta_k$ for the gap between the first conjugate-gradient iterate and the trust-region boundary, the median of ε_k is 6.536×10^{-9} at the disagreeing iterations against 1.6266×10^{-3} at the agreeing ones, five orders of magnitude apart.

The mechanism follows. At those iterations $\|z_1\|$ falls a relative 6.5×10^{-9} short of Δ_k , so (E2) does not fire and the recurrence continues. The radial room left is only about $\varepsilon_k \Delta_k$. Moving from z_1 to the boundary along any direction making an angle φ with z_1 costs a distance $\varepsilon_k \Delta_k / \cos \varphi$, so the step can rotate by at most about ε_k radians, and

$$1 - \cos(s_k, -g_k) \lesssim \frac{1}{2} \varepsilon_k^2. \quad (9)$$

The notebook measures both sides. With the median $\varepsilon_k = 6.536 \times 10^{-9}$ the bound (9) gives 2.136×10^{-17} , and the measured median of $|1 - \cos(s_k, -g_k)|$ at the disagreeing iterations is 2.220×10^{-16} , which is `eps(Float64)` exactly. The geometric bound sits below the rounding floor, so the measurement is floor limited rather than geometry limited. The measured value is four orders of magnitude below the 10^{-12} threshold of (5) and the bound is five. The Krylov dimension counts two and the angle cannot tell.

The threshold is nonetheless doing real work at the tail. The *largest* $|1 - \cos(s_k, -g_k)|$ among the disagreeing iterations is 9.2293×10^{-13} , that is 92% of 10^{-12} . The corresponding figure over the agreeing iterations has median 1.3101×10^{-6} and maximum 3.2434×10^{-1} . The two populations are separated by ten orders of magnitude in the median, and one disagreeing iteration nonetheless sits close to the boundary of the test.

Remark 1 (which one to use). **Use `cg1` as the definition of truncation and the cosine as corroboration.** The truncation condition is a statement about the conjugate-gradient recurrence, so the Krylov dimension is what states it, and the angle is a consequence of it by Proposition 1. Where they disagree, (9) shows the angle is reporting an arithmetic near-coincidence rather than a property of the step.

4 The six diagnostics

All six live in `saddle_problem.jl` and are shared by the four saddle notebooks.

`step_classes(ss)`. The triple (2) to (4), read from `:cg_iters_trajectory` and `:active_trajectory`. *Does not apply* under `ExactMS`, which reports $c_k = 0$, and returns (NaN, NaN, NaN) there rather than a fraction of zero. Read the angle instead in that case.

`eig_deviation(x)`. The sine of the angle between $H(x)g(x)$ and $g(x)$,

$$\text{eig_deviation}(x) = \sqrt{1 - \left(\frac{|(Hg)^\top g|}{\|Hg\| \|g\|} \right)^2}, \quad (10)$$

zero exactly when g is an eigenvector of H . On (1) the Hessian is diagonal on the axis $p_2 = 0$ and the gradient lies along e_1 there, so the deviation vanishes on the axis. The notebook confirms it is 0 to the last bit at $(0.3, 0)$, $(0.6, 0)$ and $(-0.5, 0)$, and 2.415×10^{-1} at the starting point. *Does not apply* where $g = 0$ or $Hg = 0$, returning NaN.

`z1_ratio(x, Δ)`. The first Steihaug iterate against the radius,

$$\text{z1_ratio}(x, \Delta) = \frac{\|z_1\|}{\Delta} = \frac{\|g\|^3}{(g^\top Hg) \Delta}. \quad (11)$$

It reaches one exactly when (E2) fires, so the truncation condition is

$$\Delta \leq \frac{\|g\|^3}{g^\top Hg}, \quad \text{equivalently} \quad \mu \leq \frac{\|g\|^2}{g^\top Hg} \quad \text{when } \Delta = \mu \|g\|, \quad (12)$$

a reciprocal curvature. A cap below that quantity truncates every step, which is the mechanism behind the top rows of Table 1. *Does not apply* where $g^\top Hg \leq 0$, returning `Inf`, because conjugate gradient then goes straight to the boundary through (E1). The notebook exhibits six such points, all near the axis between the origin and the saddle, with $g^\top Hg$ from -4.33×10^{-8} to -2.15×10^{-3} .

spectrum_variance(x). The g -weighted variance of the spectrum of $H(x)$. With $H = V\Lambda V^\top$ and weights $w_i = (V^\top g)_i^2 / \|g\|^2$,

$$\text{spectrum_variance}(x) = \sum_i w_i \lambda_i^2 - \left(\sum_i w_i \lambda_i \right)^2. \quad (13)$$

It is the coefficient in the **ExactMS** angle law of the first notebook, $1 - \cos(s_k, -g_k) \approx \frac{1}{2} \text{spectrum_variance}(\Delta_k / \|g_k\|)$. It vanishes where g is an eigenvector, so it is 0 on the axis $p_2 = 0$, which the notebook shows beside (10). *Does not apply* where $g = 0$, returning NaN.

loglog_fit(x, y). Least squares slope of $\log_{10} y$ on $\log_{10} x$, with intercept, R^2 and the number of usable points. Points with $x \leq 0$, $y \leq 0$ or a non-finite value are dropped first. *Does not apply* below five usable points, where it returns NaN, so a slope is never quoted from a handful of iterations. The notebook shows NaN at three and four points and slope 1.5 with $R^2 = 1.0$ at five and eight. **Report n and R^2 beside every slope taken from it.**

nearest_crit(x). The nearest of the four critical points and its distance, as a pair. **which_crit** is written in terms of it, returning the name when the distance is below its tolerance and "other" otherwise, so a limit point is classified by one distance computation. The notebook shows the two agreeing at the four critical points and separating at $(-0.5, 0.6)$, where the nearest is the origin at distance 7.810×10^{-1} and **which_crit** returns "other".

Never report "other" as an outcome without the distance and $\|g\|$ beside it. A run that exhausted its budget while still moving is not a run that converged somewhere else, and only those two numbers separate the cases.

5 The tables

5.1 Table 1: the degeneration ladder

Table 1: R-grad capped, truncated conjugate gradient, three model Hessians. The columns are (2) to (7). Note `cg1 = trunc + inside` on every row, to the printed precision, which is (6).

model	$\bar{\mu}$	status	iter	trunc	inside	krylov	cauchy	cg1	cos_min
ExactHessian	0.01	max_iter	2000	1.000	0.000	0.000	1.0000	1.0000	1.000000
ExactHessian	0.05	max_iter	2000	1.000	0.000	0.000	1.0000	1.0000	1.000000
ExactHessian	0.10	max_iter	2000	1.000	0.000	0.000	1.0000	1.0000	1.000000
ExactHessian	0.30	max_iter	2000	1.000	0.000	0.001	0.9995	0.9995	0.877285
ExactHessian	0.50	max_iter	2000	0.999	0.000	0.001	0.9990	0.9990	0.877285
ExactHessian	1.00	max_iter	2000	0.999	0.000	0.002	0.9985	0.9985	0.877285
ExactHessian	2.00	max_iter	2000	0.998	0.001	0.002	0.9985	0.9985	0.877285
ExactHessian	4.00	max_iter	2000	0.334	0.001	0.665	0.4885	0.3345	0.877285
ExactHessian	8.00	first_order	1678	0.331	0.001	0.668	0.4815	0.3319	0.675664
ExactHessian	128.00	first_order	85	0.047	0.035	0.918	0.0824	0.0824	0.246457
SR1	0.01	max_iter	2000	1.000	0.000	0.000	1.0000	1.0000	1.000000
SR1	0.05	max_iter	2000	1.000	0.000	0.000	1.0000	1.0000	1.000000
SR1	0.10	max_iter	2000	1.000	0.000	0.000	1.0000	1.0000	1.000000
SR1	0.30	max_iter	2000	1.000	0.000	0.000	1.0000	1.0000	1.000000
SR1	0.50	stalled	83	1.000	0.000	0.000	1.0000	1.0000	1.000000
SR1	1.00	first_order	23	0.739	0.000	0.261	0.7391	0.7391	0.671198
SR1	2.00	first_order	24	0.625	0.042	0.333	0.6667	0.6667	0.692186
SR1	4.00	first_order	19	0.632	0.105	0.263	0.7368	0.7368	0.761860
SR1	8.00	first_order	19	0.632	0.105	0.263	0.7368	0.7368	0.761860
SR1	128.00	first_order	19	0.632	0.105	0.263	0.7368	0.7368	0.761860
LBFGS	0.01	max_iter	2000	1.000	0.000	0.000	1.0000	1.0000	1.000000
LBFGS	0.05	max_iter	2000	1.000	0.000	0.000	1.0000	1.0000	1.000000
LBFGS	0.10	max_iter	2000	1.000	0.000	0.000	1.0000	1.0000	1.000000
LBFGS	0.30	max_iter	2000	1.000	0.000	0.000	1.0000	1.0000	1.000000
LBFGS	0.50	stalled	85	1.000	0.000	0.000	1.0000	1.0000	1.000000
LBFGS	1.00	first_order	30	0.333	0.100	0.567	0.4333	0.4333	0.661424
LBFGS	2.00	stalled	28	0.464	0.143	0.393	0.6071	0.6071	0.599361
LBFGS	4.00	first_order	21	0.333	0.238	0.429	0.5714	0.5714	0.426056
LBFGS	8.00	first_order	21	0.333	0.238	0.429	0.5714	0.5714	0.426056
LBFGS	128.00	first_order	21	0.333	0.238	0.429	0.5714	0.5714	0.426056

Read the table down each block. Three regimes appear, and the cap is what moves between them.

The degenerate regime, $\bar{\mu} \leq 0.1$. `trunc` = 1.000, `cauchy` = `cg1` = 1.0000 and `cos_min` = 1.000000, on all three models. Every step of two thousand was truncated at the boundary along $-g_k$. The method was gradient descent with step length Δ_k for the whole run, and `cos_min` is what says *whole*: a fraction of one leaves open a single stray iteration, and a minimum of one does not.

The mechanism is (12). A cap below $\|g_k\|^2 / (g_k^\top H_k g_k)$ truncates every step, and the model never enters. That the three models agree exactly here is the point. When the region is this small the model Hessian is irrelevant, so `SR1` and `LBFGS` produce the same run as the exact Hessian.

The mixed regime. Between $\bar{\mu} = 0.3$ and 2 under `ExactHessian`, `trunc` stays above 0.998 while `cos_min` drops to 0.877285. The fractions barely move and the minimum does, which is the

case the two kinds of column exist to separate. One or two iterations in two thousand escaped the truncated branch, and only `cos_min` records that they did.

The Krylov regime. At $\bar{\mu} = 128$ under `ExactHessian`, `krylov` = 0.918 and the run converges in 85 iterations rather than exhausting 2000. `cos_min` = 0.246457, so the step direction departed from $-g_k$ by about 76° at its worst. The model is choosing the direction, and the run is a trust-region method rather than a disguised gradient method.

The quasi-Newton models reach that regime sooner. `SR1` and `LBFGS` leave the degenerate regime at $\bar{\mu} = 1$, where `ExactHessian` is still at `trunc` = 0.999. Their runs then take 19 to 30 iterations against 2000. The cap that degenerates a method is not a property of the cap alone.

inside is almost never occupied. It is 0.000 or 0.001 on every `ExactHessian` row and reaches 0.238 only under `LBFGS`. The degeneration proceeds by truncation rather than by a short interior step, which is why `trunc` and not `cg1` is the column that identifies gradient descent.

5.2 Table 2: the two Cauchy statistics

Table 1 already shows the answer to the question of Section 3. On 28 of the 30 rows `cauchy` and `cg1` print the same number, which is Proposition 1 with an empty remainder. On the two `ExactHessian` rows at $\bar{\mu} = 4$ and 8 they separate, and the notebook’s Table 2 counts 308 and 251 disagreeing iterations, all in the direction the inclusion allows.

Table 2: The two candidate explanations, pooled over the runs that disagree at all. `eig_deviation` does not separate the two populations and the boundary gap $1 - \|z_1\|/\Delta$ separates them by five orders of magnitude.

quantity	min	median	max
<code>eig_deviation</code> , disagreeing	7.0343×10^{-1}	7.0695×10^{-1}	8.6594×10^{-1}
<code>eig_deviation</code> , agreeing	7.5345×10^{-3}	7.0617×10^{-1}	8.6594×10^{-1}
$1 - \ z_1\ /\Delta$, disagreeing	4.4409×10^{-16}	6.5360×10^{-9}	2.2238×10^{-6}
$1 - \ z_1\ /\Delta$, agreeing	-1.0790×10^0	1.6266×10^{-3}	6.8788×10^{-1}

The negative entries in the last row are not an error. On a truncating iteration $\|z_1\| \geq \Delta$, so $1 - \|z_1\|/\Delta \leq 0$, and the agreeing population is dominated by such iterations. The disagreeing population sits at $+6.5 \times 10^{-9}$, that is just *inside* the boundary, which is exactly the configuration (E2) fails to catch.

5.3 Table 3: the truncation condition, iteration by iteration

Table 3: The prediction $\|z_1\|/\Delta_k \geq 1$ against the observed $c_k = 1$. Rows with no disagreement under **SR1** and **LBFGS** are omitted; the counts below are over all ten caps.

model	$\bar{\mu}$	iterations	predicted	observed	disagreements
ExactHessian	0.01	2000	2000	2000	0
ExactHessian	0.05	2000	2000	2000	0
ExactHessian	0.10	2000	2000	2000	0
ExactHessian	0.30	2000	1999	1999	0
ExactHessian	0.50	2000	1998	1998	0
ExactHessian	1.00	2000	1997	1997	0
ExactHessian	2.00	2000	1996	1997	1
ExactHessian	4.00	2000	667	669	2
ExactHessian	8.00	1678	555	557	2
ExactHessian	128.00	85	4	7	3
SR1	0.30	2000	1999	2000	1
SR1	0.50	82	81	82	1
SR1	1.00	23	16	17	1
SR1	2.00	24	14	16	2
SR1	4.00	19	11	14	3
LBFGS	0.30	2000	1999	2000	1
LBFGS	0.50	84	83	84	1
LBFGS	1.00	30	19	13	12
LBFGS	2.00	27	16	16	4
LBFGS	4.00	21	11	12	5

Under **ExactHessian** the prediction is right on 17755 of 17763 iterations, a disagreement rate of 0.0005. The rate is 0.0017 under **SR1** and 0.0040 under **LBFGS**, and that difference is a property of the diagnostic rather than of the condition: **z1_ratio** reads the *true* Hessian through **hess**, and conjugate gradient runs on the *model* Hessian. The two coincide under **ExactHessian** and nowhere else.

Table 4: Every disagreement under **ExactHessian**, individually. Three distinct iterates account for all eight, appearing once per cap large enough to reach them.

$\bar{\mu}$	k	z1_ratio	c_k	eig-deviation	Δ_k	$\ g_k\ $
2.00	3	0.74290	1	7.5345e-03	6.1095e-02	3.0547e-02
4.00	3	0.74290	1	7.5345e-03	6.1095e-02	3.0547e-02
4.00	4	0.76071	1	5.3537e-02	3.1586e-02	7.8966e-03
8.00	3	0.74290	1	7.5345e-03	6.1095e-02	3.0547e-02
8.00	4	0.76071	1	5.3537e-02	3.1586e-02	7.8966e-03
128.00	3	0.74290	1	7.5345e-03	6.1095e-02	3.0547e-02
128.00	4	0.76071	1	5.3537e-02	3.1586e-02	7.8966e-03
128.00	7	0.78195	1	9.8707e-03	4.1855e-03	1.3080e-04

All eight have the same shape. **z1_ratio** lies between 0.7429 and 0.7819, so the prediction says the step will not truncate, and $c_k = 1$ says it stopped after one iteration anyway. None is within 10^{-6} of one and none is **Inf**, so neither (E2) nor (E1) explains them.

The cause is (E3). **z1_ratio** models the boundary exit and nothing else, while the recurrence has three exits. Those eight iterations stopped *inside* the region because the residual met the forcing condition, which (11) cannot see.

Remark 2 (the prediction is one-sided, and that is what to rely on). There is no disagreement in the other direction. Over all 17763 **ExactHessian** iterations, **z1_ratio** ≥ 1 never occurs with

$c_k \geq 2$. So $\|z_1\| / \Delta_k \geq 1$ is *sufficient* for truncation and not necessary, which is what (12) asserts and all that Section 5.1 needs from it.

5.4 What the four tables say together

The cap controls the radius, the radius controls the truncation through (12), and the truncation controls whether the model Hessian influences the step at all. Table 1 measures the last of those three, Table 3 confirms the middle one to a rate of five in ten thousand, and Tables 2 and 4 account for every iteration where the diagnostics disagree.

Two of the eight columns are load bearing and the rest corroborate. **trunc** identifies gradient descent in disguise, because it is the only class in which neither the direction nor the length came from the model. **cos_min** says whether that held throughout the run or only on average. **cauchy**, **cg1**, **inside** and **krylov** are consistent with those two by (6) and Proposition 1, and where they part company the reason is arithmetic rather than algorithmic.